

SEMESTER END EXAMINATIONS – MARCH 2024

Program	: Science / Artificial Intelligence and Machine Learning	Semester	: III
Course Name	: Introduction to Artificial Intelligence	Max. Marks	: 100
Course Code	: AD35 / AI35	Duration	: 3 Hrs

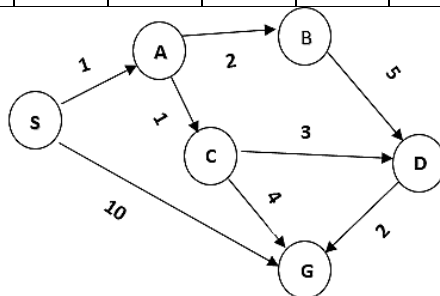
Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Describe model-based reflex agent and utility-based agent with neat diagrams. CO1 (08)
 - Write and explain the Breadth-first search algorithm. CO1 (08)
 - Illustrate with an example Acting Humanly approach to Artificial Intelligence. CO1 (04)
- Differentiate between following properties of task environments with an example. CO1 (06)
 - Fully Observable Vs. Partially Observable
 - Static vs Dynamic.
 - What is PEAS in AI? Illustrate the PEAS description for the following agent types. CO1 (06)
 - Automated taxi driver
 - Part – picking robot.
 - Solve the following using A* Search CO1 (08)

State	S	A	B	C	D	G
H(n)	5	3	4	2	6	0



UNIT - II

- Write the sequence of steps required to grab the gold in typical wumpus world. CO2 (05)
 - Write the predicate logic for the following sentences: CO2 (06)
 - Richard, the Lionheart's father is married to King John's mother
 - There is at least one student who does not like homework.
 - King John has a crown on his head.
 - Discuss the steps used in knowledge engineering process using first-order logic. CO2 (09)

4. a) Illustrate forward-chaining algorithm using predicate logic. CO2 (09)
b) Convert following sentences into first order logic: CO2 (06)
i. Not all students like both maths and science
ii. An OR gate's output is 1 if and only if any of its inputs is 1
iii. If two terminals are connected, then they have the same signal.
c) Discuss in brief propositional vs. First-Order Inference. CO2 (05)

UNIT - III

5. a) Discuss five basic probability notation with propositional logic. CO3 (06)
b) Explain with an example text classification using naive Bayes. CO3 (08)
c) Illustrate boosting ensemble model with an example. CO3 (06)
6. a) Write a PDDL description for the spare tire problem in classical planning. CO3 (06)
b) Illustrate support vector machine model with an example. CO3 (06)
c) Explain the following two approaches of classical planning in AI with example. CO3 (08)
i. Forward (progression) search through the space of ground states.
ii. Backward (regression) search through state descriptions.

UNIT- IV

7. a) Explain how the process of text classification is handled in NLP. Explain the process of classifying an email as a spam or not-spam (Ham), with an example. CO4 (06)
b) Explain with an example on hot vector and word embeddings in NLP. CO4 (06)
c) Illustrate a detailed note on pretrained contextual representations. CO4 (08)
8. a) Explain with an example semantic interpretation in natural language processing. CO4 (06)
b) Make use of a suitable example to describe pretrained contextual presentation. CO4 (06)
c) Illustrate the usage of Recurrent Neural Networks for Natural Language Processing. CO4 (08)

UNIT - V

9. a) Categorize Robots based on their types, Manipulation, environment and the device it's made of. Give examples for each type. CO5 (08)
b) Discuss the Significance of genetic operators. CO5 (08)
c) Outline the safety and limits of the AI. CO5 (04)
10. a) With suitable example, discuss how reinforcement learning helps to improve the robot ability to learn, improve, adapt and reproduce tasks with dynamically changing constraints. CO5 (08)
b) Illustrate hypothesis representation in genetic algorithm. CO5 (08)
c) Describe how robotic technology is used in various applications. CO5 (04)
